Operating system

An interface b/w the user and that manages computer resources provides services for program.

1. OS Types.

1. Batch OS
2. Multiprogramming OS
3. Multitasking OS
4. Multiprocessing OS
5. Distributed OS
6. Network OS
7. Real-time OS (Hard/soft)

2. OS structure

1) Simple structure
2) Layered structure
3) Microkernal structure
4) Monolithic Kernal structure
5) Hybrid kernal structure

4. OS Functions

1. Process Management
2. Memory Management
3. File Management
4. Device Management
5. CPU scheduling
6. I/O Management
7. security & protection

3. system calls

1. Process control
* fork()
* exec()
* exit()
* wait()

2. File Management
* open()
* close()
* read()
* write()
* create()

3. Device Management
* read()
* write()
* ioct()
* request device

4. information maintence
* getpid()
* time()
* unname()
* get /set

5. communication
* pipe()
* socket()
* send()
* receive()
* shared memory()

5. Cross-links

OS Type influences choice of OS structure

Network OS & Distributed OS depend heavily on communication

OS structure determine how system call are implemented

Microkernal structure more system calls via message passing

system calls provide access to OS function

OS Functions support all OS Types

process Management uses CPU scheduling for execution.